

STAT452

Polynomial Regression and transformation of variables

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Outline

- 1 Introduction
- 2 Considerations in fitting polynomial in one variable
- 3 Example
- 4 Transformation of Variables

Non-linear effects of predictors – Polynomial Regression

- The relationship between Y and X often turns out not to be a straight line.
- How do we assess the significance of departures from linearity?
- One of the simplest ways is to use *polynomial regression*.
- As before, we have just *one continuous explanatory variable*, X , but we can fit *higher powers* of X , such as X^2 and X^3 , to the model in addition to X to explain curvature in the relationship between Y and X .

Polynomial Regression

We can use **polynomial regression** models such as

$$y = \beta_0 + \beta_1 x + \cdots + \beta_k x^k + \epsilon$$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \beta_{12} x_1 x_2 + \epsilon$$

to represent the nonlinear patterns

- The top model is called a **kth-order polynomial model in one variable**;
- The bottom model is called a **quadratic model in two variables**.

Polynomial Regression

Polynomial models are very powerful to handle nonlinearity, because **polynomials can approximate continuous functions within any given precision**.

However, such fitting problems can still be treated as **linear regression**

$$y = \beta_0 + \beta_1 \underbrace{x}_{x_1} + \cdots + \beta_k \underbrace{x^k}_{x_k} + \epsilon,$$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \underbrace{\beta_{11} x_1^2}_{\beta_3 x_3} + \underbrace{\beta_{22} x_2^2}_{\beta_4 x_4} + \underbrace{\beta_{12} x_1 x_2}_{\beta_5 x_5} + \epsilon.$$

Thus, we can readily utilize the corresponding theory, tools and techniques for linear regression to carry out polynomial regression.

Matrix representation of polynomial regression

Consider the polynomial regression model of order 2:

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \varepsilon, \quad i = 1, \dots, n.$$

The matrix form of the problem is

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix},$$

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2)'$.

Important considerations

There will be important considerations in polynomial regression:

- **Order of the polynomial model**
- **Model-building strategy**
- **Extrapolation**
- **Ill-conditioning**
- **Hierarchy**

Order of the polynomial model

- It is always possible to fit a polynomial model of order $n - 1$ perfectly to a data set n points (however, this will almost surely be overfitting!!!)
- Transformations should be tried first to keep the model first order.
- A low-order model in a transformed variable is almost always preferable to a high-order model in the original metric.
- One should always maintain a sense of **parsimony**, that is, **use the simplest possible model** that is consistent with the data and knowledge of the problem environment.

Model-building strategy

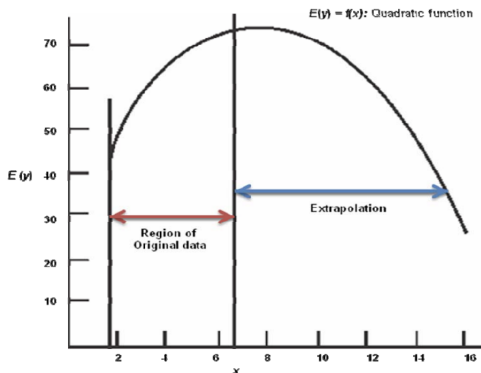
There are two standard procedures for building a polynomial model:

- **Forward selection:** Successively fit models of increasing order until the t test for the highest order term is nonsignificant.
- **Backward elimination:** Appropriately fit the highest order model and then delete terms one at a time, starting with the highest order, until the highest order remaining term has a significant t statistic.

The forward selection and backward elimination procedures do not necessarily lead to the same model. The first and second-order polynomials are mostly used in practice.

Extrapolation

- One has to be very cautioned in extrapolation with polynomial models. The curvatures in the region of data and the region of extrapolation can be different.



Ill-conditioning

- In the setting of polynomial regression, the design matrix may have **lots of columns** corresponding to just one predictor or two.
- Those columns will have a **significant multicollinearity**, especially when the values of x are limited to a narrow range.
- **As the order of the polynomial model increases, $\mathbf{X}'\mathbf{X}$ become more and more ill-conditioned**, meaning that matrix inversion calculations are more and more inaccurate.
- **Centering the data (i.e., letting $\tilde{x}_i = x_i - \bar{x}$) may remove some nonessential ill-conditioning.**

Hierarchy

- A model is said to be hierarchical if it contains the terms x , x^2 , x^3 , etc. in a hierarchy. For example, the model

$$y = \beta_0 + \beta_1x + \beta_2x^2 + \beta_3x^3 + \beta_4x^4 + \varepsilon$$

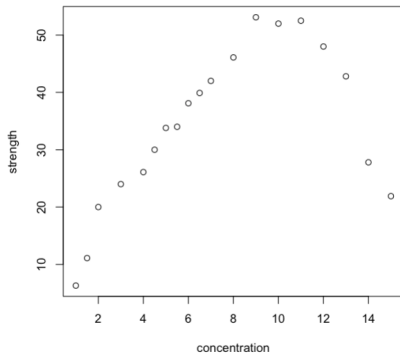
is hierarchical as it contains all the terms up to order four.

- The model $y = \beta_0 + \beta_1x + \beta_2x^2 + \beta_4x^4 + \varepsilon$ is not hierarchical as it does not contain the term of x^3 .
- It is expected that all polynomial models should have this property because only hierarchical models are invariant under linear transformation.

Example: The Hardwood Data

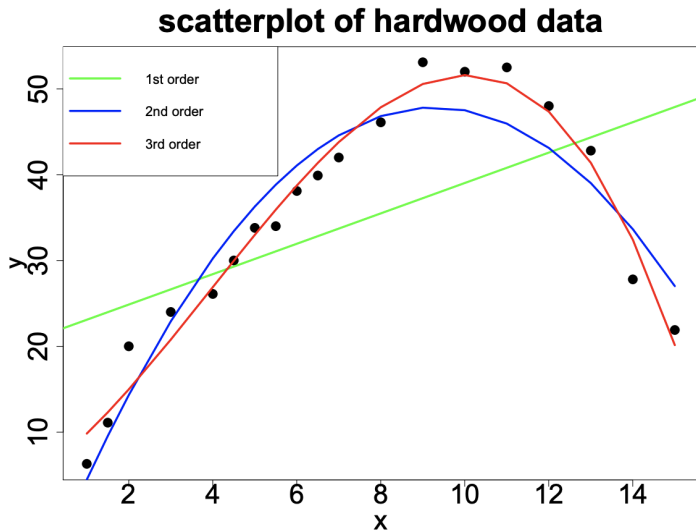
We have 19 observations concerning the **strength of kraft paper** (y) and the **percentage of hardwood** (x) in the batch of pulp from which the paper was produced.

Three polynomial models along with the linear model were fitted to the data.



Scatterplot of hardwood data

Example: The Hardwood Data



Example 1: The Hardwood Data

The summary statistics of the four models are reported below:

Order of poly. model	R^2	R_{adj}^2	$\hat{\sigma}^2$	p-value*
1 (linear model)	0.3054	0.2645	11.82^2	0.01414
2 (quadratic model)	0.9085	0.8971	4.42^2	1.89×10^{-8}
3 (cubic model)	0.9707	0.9648	2.585^2	4.72×10^{-5}
4	0.9736	0.9661	2.539^2	0.233

*of the t test for the highest-order term in each model.

Which model should we select?

Multicollinearity: Variance Inflation Factor (VIF)

- VIF measures the degree of multicollinearity among predictor variables in a regression model.
- Formula:

$$VIF_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is the coefficient of determination when regressing variable X_j on the other predictors.

- Rule of thumb:
 - $VIF = 1$: no multicollinearity,
 - $VIF > 5$: moderate multicollinearity,
 - $VIF > 10$: severe multicollinearity.

Example 1: The Hardwood Data - VIF

Model: $y \sim x + x^2$

Variable	x	x^2
VIF	1.09705	1.09705

- Very low VIF values indicate negligible multicollinearity.
- The model is stable with independent predictors.

Model: $y \sim x + x^2 + x^3$

Variable	x	x^2	x^3
VIF	6.13	1.29	6.82

- Multicollinearity starts to emerge, especially between x and x^3 .
- Still usable, but results should be interpreted with caution.

Example 1: The Hardwood Data - VIF

Model: $y \sim x + x^2 + x^3 + x^4$

Variable	x	x^2	x^3	x^4
VIF	7.80	14.71	11.49	19.58

- Severe multicollinearity observed.
- Coefficients may become unstable and hard to interpret.
- Consider alternative approaches.

Multicollinearity: recommendations

- Use `poly(x, degree=n)` in R to create orthogonal polynomial terms.
- Consider standardizing predictors before polynomial transformation.
- Try regularized regression (e.g., Ridge or Lasso) when using high-degree polynomials.
- Always check VIF to monitor multicollinearity risk.

Interpretation of Coefficients in a Polynomial Model

Recall in previous lecture, we said β_j = the regression coefficient for x_j , is the mean change in the response y when x_j is increased by one unit **holding other x_i 's constant**.

However, in a model that involves polynomial terms like

$$y = \beta_0 + \underbrace{\beta_1 x_1 + \beta_2 x_1^2}_{\text{a polynomial of } x_1} + \beta_3 x_3 + \cdots + \beta_p x_p + \epsilon$$

it makes no sense to interpret a single coefficient for a polynomial like β_1 or β_2 since it's impossible to change x_1 while holding x_1^2 constant.

We should interpret the entire polynomial altogether, like:

*the mean of y change with x_1 following the curve $\beta_1 x_1 + \beta_2 x_1^2$ **holding other x_i 's constant**.*

Why Do We Transform Variables?

Transforming variables—either predictors or responses—is a fundamental technique in regression modeling, typically used to address the following issues:

- **Non-linearity:** When the relationship between predictors and the response is not linear, transformation can help achieve linearity.
- **Non-constant variance (heteroscedasticity):** Transformation stabilizes variance across observations, enabling better model assumptions.
 - Common techniques include:
 - Variance-stabilizing transformations (e.g., log, square root)
 - Box-Cox transformation family

Linear vs. Nonlinear Models

It is important to distinguish between models that are **linear in parameters** and those that are not.

Examples of linear models (despite non-linear terms in predictors):

- $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon$
- $Y = \beta_0 + \beta_1 \log(X) + \varepsilon$
- $Y = \beta_0 + \beta_1 \sqrt{X} + \varepsilon$

These are linear because parameters $\beta_0, \beta_1, \beta_2$ enter the model linearly.

Nonlinear model example:

$$Y = \beta_0 + \exp(\beta_1 X) + \varepsilon$$

This is *nonlinear in parameters* and cannot be handled using standard linear regression techniques without transformation.

Transforming Nonlinear Models into Linear Ones

Many nonlinear relationships can be made linear through appropriate transformations of variables.

Example 1: Exponential Growth/Decay

$$Y = \alpha e^{\beta X}$$

Taking logarithms:

$$\log(Y) = \log(\alpha) + \beta X$$

This yields a linear relationship between $\log(Y)$ and X .

Example 2: Learning Curve in Psychology

$$T_i = \alpha \beta^i, \quad \alpha > 0, \quad 0 < \beta < 1$$

Taking logarithms:

$$\log(T_i) = \log(\alpha) + \log(\beta)i$$

Now the model becomes linear in i .

Examples of Linearizable Models

Model Form	Transformation	Linear Form
$Y = \alpha X^\beta$	$\log Y, \log X$	$\log Y = \log \alpha + \beta \log X$
$Y = \alpha e^{\beta X}$	$\log Y$	$\log Y = \log \alpha + \beta X$
$Y = \alpha + \beta \log X$	$\log X$	$Y = \alpha + \beta \log X$
$Y = \frac{X}{\alpha X - \beta}$	$1/Y, 1/X$	$1/Y = \alpha - \beta(1/X)$
$Y = \frac{e^{\alpha + \beta X}}{1 + e^{\alpha + \beta X}}$	$\log \frac{Y}{1-Y}$	$\log \frac{Y}{1-Y} = \alpha + \beta X$

These transformations allow us to apply standard multiple linear regression (MLR) techniques, even when the original model is nonlinear.

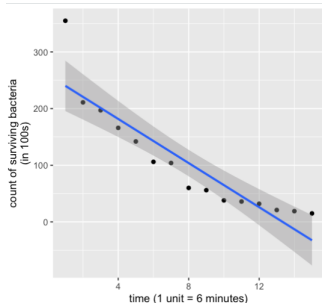
Example: Bacteria Deaths Due to X-Ray Radiation

- t : time index (1 unit = 6 minutes),
- $N_t = n_t$: Number of surviving bacteria (in hundreds) at time t after exposure to 200-kilovolt X-rays.

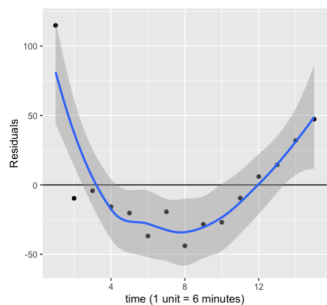
First attempt: Fit a simple linear regression (SLR) model:

$$\text{lm1} = \text{lm}(N_t \sim t, \text{data} = \text{bact}).$$

We then plot the scatterplot and the residual plot.

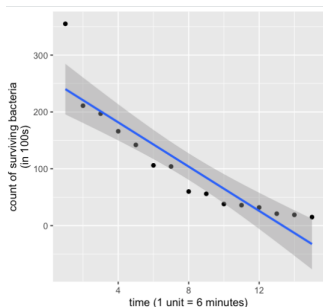


Scatterplot: N_t vs. time.

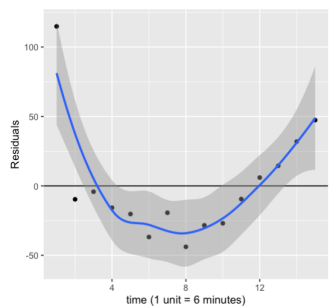


Residual plot from linear fit.

Diagnosing Non-Linearity via Plots



Scatterplot: N_t vs. time.



Residual plot from linear fit.

When deciding on appropriate transformations, it is essential to begin with visual diagnostics:

- The scatterplot reveals an **overall downward trend**, but subtle non-linearity may be hard to detect.
- The residual plot clearly shows a **curved pattern**, indicating that the linear model does not adequately capture the relationship.
- In practice, non-linearity may not always be obvious in the scatterplot—**always check residuals**.

Example: Bacteria Deaths Due to X-Ray Radiation

Based on biological theory, we expect the number of surviving bacteria to decay exponentially over time:

$$N_t = N_0 e^{\beta_1 t}, \quad \text{where} \quad \begin{cases} N_0 & : \text{initial population size} \\ \beta_1 & : \text{decay rate} \end{cases}$$

Taking the natural logarithm on both sides yields:

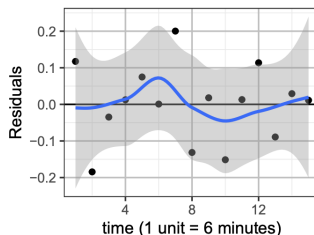
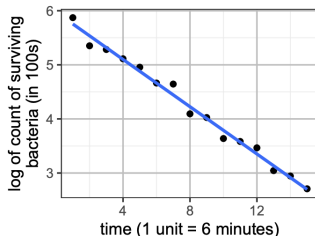
$$\log N_t = \log N_0 + \beta_1 t = \beta_0 + \beta_1 t,$$

where $\beta_0 = \log N_0$.

This suggests a linear relationship between $\log N_t$ and t . Hence, we can fit the model using:

$$\text{lm1} = \text{lm}(\log(N_t) \sim t, \text{data} = \text{bact}).$$

Evaluating the Log-Transformed Model



- The scatterplot shows that applying a log transformation results in a clearly linear pattern.
- The residual plot reveals no obvious structure, suggesting that:
 - Linearity is achieved.
 - Homoscedasticity and independence assumptions are reasonably satisfied.
- Thus, the transformed model is a better fit for the data.

Interpretation of the Exponential Decay Model

Consider the exponential model:

$$N_t = N_0 e^{\beta_1 t}.$$

Each additional time unit (6 minutes) results in the bacterial count being scaled by e^{β_1} .

Based on the fitted model:

```
lm2 = lm(log(N_t) ~ t, data = bact)
summary(lm2)$coef
```

Estimated coefficient for time: $\hat{\beta}_1 = -0.2184$

$$\text{Proportional change} = 1 - e^{-0.2184} \approx 19.6\%.$$

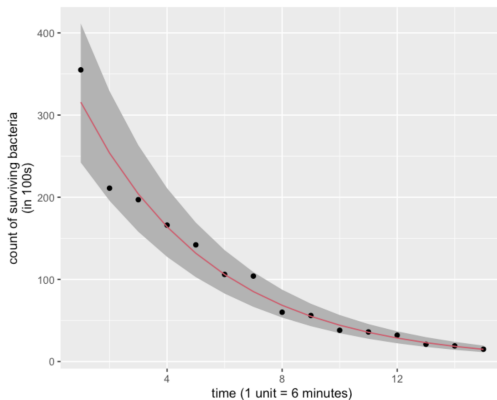
Thus, the number of surviving bacteria decreases by approximately **19.6% every 6 minutes**.

95% CI:

$$1 - e^{-0.2326} \approx 18.5\% \quad \text{to} \quad 1 - e^{-0.2042} \approx 20.8\%.$$

Back to the Original Scale

Step: Exponentiate predicted values from the log-linear model to obtain predictions on the original scale.



The transformed model captures the exponential decay pattern well when converted back to the original scale.

Logarithm: A Widely Used Transformation

- When the error term is multiplicative and proportional to the mean:

$$Y = f(X_1, \dots, X_p)(1 + \varepsilon) \Rightarrow \log Y = \log f(X_1, \dots, X_p) + \log(1 + \varepsilon).$$

- **Rule of thumb #1:** Use log when the variable reflects an **amount of money**.
 - e.g., Per capita income, prices, salaries.
- **Rule of thumb #2:** Use log for variables measuring **concentration** or **rate**.
 - e.g., chemical concentration in blood, population density.
- **Rule of thumb #3:** If a variable spans several orders of magnitude (e.g. from 10 to 10,000), apply log to compress scale and stabilize variance.

Interpreting Log-Transformed Variables

1. Semi-log model:

$$\log(Y) = \beta_0 + \beta_1 X \Rightarrow Y = e^{\beta_0} e^{\beta_1 X}.$$

- Interpretation: Each 1-unit increase in X multiplies Y by e^{β_1} .

2. Log-log model:

$$\log(Y) = \beta_0 + \beta_1 \log(X) \Rightarrow Y = e^{\beta_0} X^{\beta_1}.$$

- When X doubles, Y is multiplied by 2^{β_1} .
- In economics, β_1 is known as the **elasticity**:

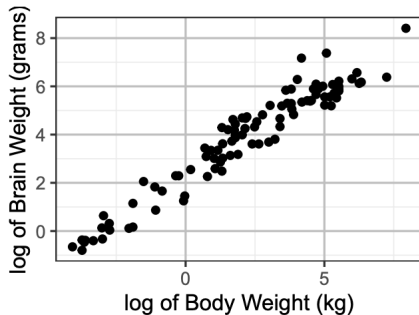
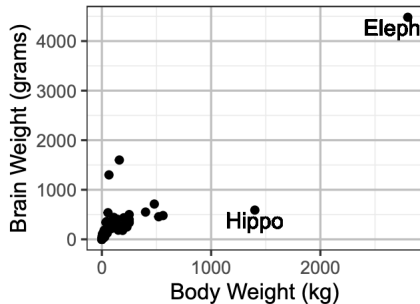
$$\frac{dY}{Y} = \beta_1 \frac{dX}{X}.$$

- So a 1% increase in X leads to a $\beta_1\%$ increase in Y .

Transformations to Reduce Skewness

- **Skewed response variables** may violate the normality assumption of the error term ε .
 - Mild non-normality is often tolerable if there are no major issues with non-linearity or heteroscedasticity.
 - In such cases, transformation may not be necessary.
- **Skewed predictors** can lead to:
 - Extreme outliers that distort model fit.
 - Undue influence from high-leverage points.
- **Transforming variables** (e.g., log, square root) can:
 - Stabilize variance and improve normality.
 - Reduce the impact of extreme values and improve model robustness.
- *Tip:* If you detect outliers or skewness, consider applying transformations before modeling.

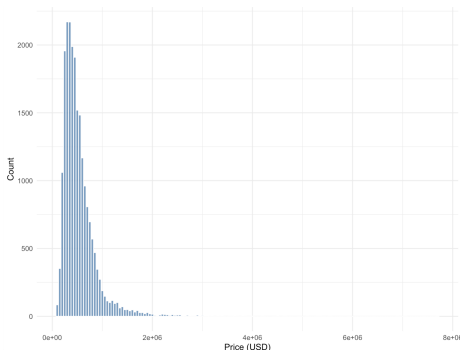
Example: Brain and Body Weight of Mammals



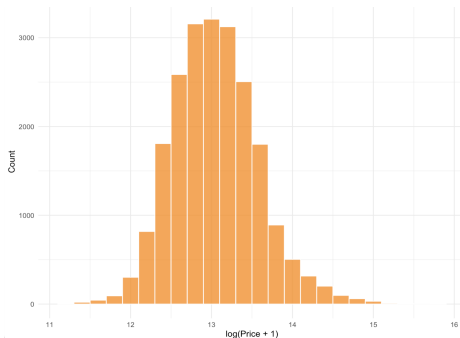
Before transformation, both Brain weight and Body weight are highly right-skewed.

Reducing Skewness with Log Transformation: Housing Prices

- Housing prices are often highly **right-skewed** due to a small number of very expensive properties.
- Applying a log transformation to the price variable compresses the upper tail, yielding a more symmetric distribution.



Original price



log(price)

Transformations to Reduce Skewness

Skewness in a variable can often be reduced using a **power transformation**, defined as:

$$f_{\alpha}(y) = \begin{cases} y^{\alpha}, & \text{if } \alpha \neq 0 \\ \log(y), & \text{if } \alpha = 0. \end{cases}$$

- **For right-skewed distributions**, consider transformations with $\alpha < 1$, such as:

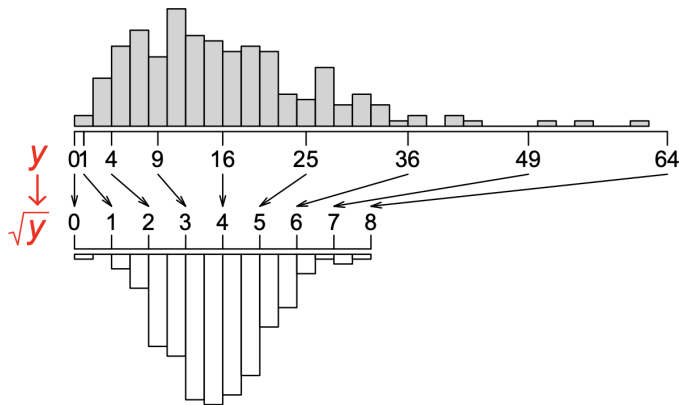
$$y \longrightarrow \frac{1}{y}, \log(y), \sqrt{y}, \text{ or } y^{\alpha} \text{ with } \alpha < 1$$

- **For left-skewed distributions**, use transformations with $\alpha > 1$, including:

$$y \longrightarrow y^2, y^3, \text{ or } y^{\alpha} \text{ with } \alpha > 1$$

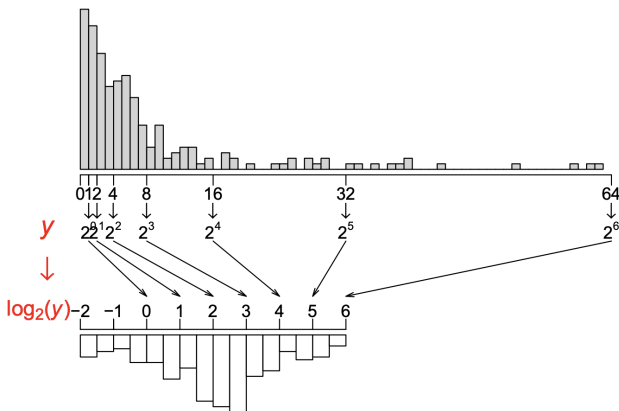
Square-Root Transformation Reduces Right-Skewness

The square-root transformation reduces skewness by compressing larger values more than smaller ones. This effect shortens the right tail and stretches the lower end of the distribution, leading to a more symmetric shape.



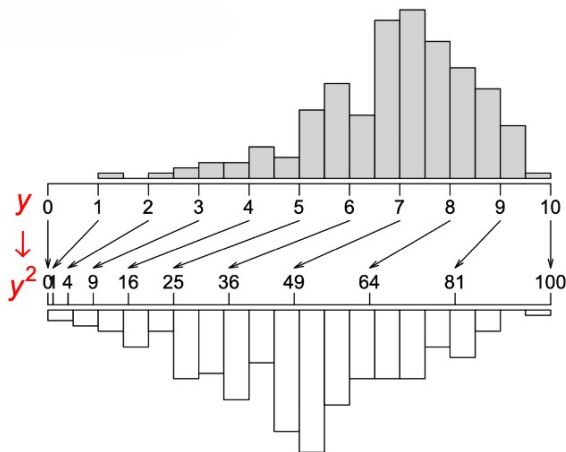
Log Transformation Further Reduces Right-Skewness

The logarithmic transformation (e.g., using base 2 or the natural logarithm) provides even greater compression of large values compared to the square-root transformation. This makes it especially effective for highly skewed data or variables spanning several orders of magnitude.



Square Transformation Reduces Left-Skewness

The square transformation increases the spread of larger values more than smaller ones. As a result, it stretches the upper tail and compresses the lower end of the distribution, thereby reducing left-skewness— though it may introduce some degree of right-skewness.



Caution When Transforming Variables

- Transformations are powerful tools that can make a linear regression model more appropriate by rescaling or re-expressing variables.
- However, **transformations are not a cure-all**. They may not fully resolve issues such as:
 - Non-linearity in the relationship between variables — in which case a non-linear model may be more suitable.
 - Non-constant variability (heteroscedasticity) — where **weighted least squares** may be a better approach if no effective transformation is available.

Caution When Transforming Variables

- Transformed variables can be more difficult to interpret, especially for non-technical audiences.
- There are often multiple ways to transform a variable, and rarely a single “correct” one. It’s common to experiment with different transformations and select the one that balances model fit and interpretability.
- **Important:** When you transform a variable, all model estimates and confidence intervals are in the transformed scale. To report results meaningfully, you must convert them **back to the original scale**.